

The Ornstein–Uhlenbeck Process: Exact Solution and Moments

hasklib mathematical companion

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Abstract

This note derives the result `OU::sample_terminal` relies on: the Ornstein–Uhlenbeck (OU) SDE is linear enough that Itô’s lemma applied to the integrating factor $e^{\kappa t} X_t$ removes the X_t -dependence entirely, giving an exact Gaussian terminal law rather than requiring discretisation. We derive that law, its mean and variance, and the stationary distribution as $t \rightarrow \infty$. This file assumes Itô’s lemma as established in `ito_foundations.tex` (Theorem 1 there) and a standard fact about Wiener integrals (the Itô isometry, cited below); neither is re-derived here.

1 The OU SDE

Definition 1 (Ornstein–Uhlenbeck process). X_t follows an Ornstein–Uhlenbeck process with mean-reversion speed $\kappa > 0$, long-run mean $\theta \in \mathbb{R}$, and volatility $\sigma > 0$ if it solves

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t, \quad X_0 \in \mathbb{R}. \quad (1)$$

In the notation of `ito_foundations.tex` this is the choice $a(t, x) = \kappa(\theta - x)$, $b(t, x) = \sigma$ (constant) in the general SDE $dX_t = a(t, X_t)dt + b(t, X_t)dW_t$.

Remark 1 (Correspondence with the code). `OU::drift` returns $\kappa(\theta - x)$ and `OU::diffusion` returns the constant σ , independent of x — unlike GBM, the diffusion here does not scale with the state. `EulerMaruyama` can still step this process through the same `StochasticProcess` interface, but as with GBM there is an exact alternative, derived below, that `OU::sample_terminal` uses instead.

Both coefficients are affine (in particular globally Lipschitz and of linear growth) in x , so the existence–uniqueness assumption of `ito_foundations.tex` applies unconditionally here — unlike GBM, there is no positivity constraint to track, since OU is free to take any real value.

2 Exact solution: the integrating-factor trick

Unlike GBM, taking a logarithm does nothing useful here since (1) is already linear in X_t , not multiplicative. The relevant substitution instead is the same one used for linear ODEs: multiply by the integrating factor $e^{\kappa t}$ and let Itô’s lemma absorb the product rule.

Theorem 1 (Exact terminal law of OU). *Let X_t solve (1). Then for any $T > 0$,*

$$X_T = X_0 e^{-\kappa T} + \theta(1 - e^{-\kappa T}) + \sigma e^{-\kappa T} \int_0^T e^{\kappa s} dW_s. \quad (2)$$

Conditional on X_0 , X_T is Gaussian.

Proof. Apply Itô's lemma (Theorem 1 of `ito_foundations.tex`) to $f(t, x) = e^{\kappa t} x$, so $f_t = \kappa e^{\kappa t} x$, $f_x = e^{\kappa t}$, $f_{xx} = 0$. With $a(t, X_t) = \kappa(\theta - X_t)$ and $b(t, X_t) = \sigma$,

$$d(e^{\kappa t} X_t) = \left(\kappa e^{\kappa t} X_t + \kappa(\theta - X_t) e^{\kappa t} + 0 \right) dt + \sigma e^{\kappa t} dW_t.$$

The X_t terms cancel completely — $\kappa e^{\kappa t} X_t$ against $-\kappa X_t e^{\kappa t}$ — which is precisely the point of the integrating factor, leaving

$$d(e^{\kappa t} X_t) = \kappa \theta e^{\kappa t} dt + \sigma e^{\kappa t} dW_t. \quad (3)$$

The right-hand side no longer involves X_t at all, so (3) integrates directly from 0 to T :

$$e^{\kappa T} X_T - X_0 = \kappa \theta \int_0^T e^{\kappa s} ds + \sigma \int_0^T e^{\kappa s} dW_s = \theta(e^{\kappa T} - 1) + \sigma \int_0^T e^{\kappa s} dW_s.$$

Multiplying by $e^{-\kappa T}$ gives (2). The stochastic integral $\int_0^T e^{\kappa s} dW_s$ is a Wiener integral of a deterministic integrand, hence Gaussian (a linear functional of a Gaussian process); X_T is an affine transformation of it and of the deterministic terms, so X_T is Gaussian as well. $\square$

Lemma 1 (Moments of a Wiener integral). *For deterministic $g \in L^2[0, T]$, the stochastic integral $I = \int_0^T g(s) dW_s$ satisfies $\mathbb{E}[I] = 0$ and $\text{Var}(I) = \int_0^T g(s)^2 ds$ (the Itô isometry).*

Citation. This is a standard property of the Itô integral, following from its construction as an L^2 -limit of Riemann-type sums of independent Gaussian increments; see Shreve (2004) or Øksendal (2003). (*Cited, not proved, in the same spirit as the existence–uniqueness assumption of `ito_foundations.tex` — the library uses the conclusion, not the construction.*) $\square$

3 Moments and the stationary law

Proposition 1 (Mean and variance of OU).

$$\mathbb{E}[X_T | X_0] = X_0 e^{-\kappa T} + \theta(1 - e^{-\kappa T}), \quad \text{Var}(X_T | X_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa T}).$$

Proof. Take expectations in (2). By lemma 1 with $g(s) = \sigma e^{-\kappa T} e^{\kappa s}$ (constant $e^{-\kappa T}$ pulled outside the integral), $\mathbb{E}[\int_0^T e^{\kappa s} dW_s] = 0$, so the mean is just the deterministic part:

$$\mathbb{E}[X_T | X_0] = X_0 e^{-\kappa T} + \theta(1 - e^{-\kappa T}).$$

For the variance, only the stochastic term contributes (the rest of (2) is deterministic given X_0):

$$\text{Var}(X_T | X_0) = \sigma^2 e^{-2\kappa T} \text{Var}\left(\int_0^T e^{\kappa s} dW_s\right) = \sigma^2 e^{-2\kappa T} \int_0^T e^{2\kappa s} ds,$$

using lemma 1 with $g(s) = e^{\kappa s}$. The integral evaluates to $(e^{2\kappa T} - 1)/(2\kappa)$, giving

$$\text{Var}(X_T | X_0) = \sigma^2 e^{-2\kappa T} \cdot \frac{e^{2\kappa T} - 1}{2\kappa} = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa T}). \quad \square$$

Remark 2. Both the mean and the variance are exactly what `OU::sample_terminal` needs: draw $Z \sim \mathcal{N}(0, 1)$ once and set

$$X_T = X_0 e^{-\kappa T} + \theta(1 - e^{-\kappa T}) + \sqrt{\frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa T})} Z,$$

exactly as GBM's exact sampler needed only Z and the closed-form mean and variance from `gbm.tex`. No time-stepping, hence no discretisation bias.

Corollary 1 (Stationary law). *As $T \rightarrow \infty$ (with $\kappa > 0$ fixed), $X_T \mid X_0$ converges in distribution to*

$$X_\infty \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{2\kappa}\right),$$

independent of the starting point X_0 .

Proof. Immediate from proposition 1: $e^{-\kappa T} \rightarrow 0$ and $e^{-2\kappa T} \rightarrow 0$ as $T \rightarrow \infty$ since $\kappa > 0$, so the mean $\rightarrow \theta$ and the variance $\rightarrow \sigma^2/(2\kappa)$, independent of X_0 . $\square$

Remark 3 (Why $\kappa > 0$ matters). The mean-reversion speed $\kappa > 0$ is what makes $e^{-\kappa T} \rightarrow 0$ and hence guarantees a stationary law at all; it is also what makes OU a sensible model for a quantity that fluctuates around a long-run level θ rather than drifting away from it, in contrast to GBM's persistent (possibly explosive) drift. If $\kappa \leq 0$ the process is non-stationary and corollary 1 does not apply — the library assumes $\kappa > 0$ throughout.

Remark 4 (Relevance beyond the pricing showcase). OU is not only one of the four processes in the convergence demonstration; it is also the driving process behind the Hull–White short-rate model, $dr_t = \kappa(\theta(t) - r_t)dt + \sigma dW_t$, used throughout Brigo & Mercurio (2006) for term-structure modelling. The only difference from (1) is a time-dependent long-run level $\theta(t)$ calibrated to the observed yield curve, chosen precisely so that the affine structure exploited in theorem 1 survives with θ replaced by $\theta(t)$ inside the integral. This is the model relevant to the callable-note decomposition (cancelable payer swap plus receiver swaption) discussed elsewhere in this thesis.

References

- Brigo, D. & Mercurio, F. (2006), *Interest Rate Models: Theory and Practice*, Springer Finance, 2nd edn, Springer, Berlin.
- Øksendal, B. (2003), *Stochastic Differential Equations: An Introduction with Applications*, 6th edn, Springer, Berlin.
- Shreve, S. E. (2004), *Stochastic Calculus for Finance II: Continuous-Time Models*, Springer Finance, Springer, New York.